Organization Data Filtering and Analysis using SQL

Introduction

In this project, I used SQL queries to extract and analyse information from an organization's database within a virtual machine (VM) environment. The goal was to practice retrieving and filtering specific data using the WHERE and LIKE operators.

The project involved exploring multiple tables within the database, including employee and device information. I ran SQL queries to identify details such as device IDs, operating systems, and operating system versions. Additionally, I filtered employee data to determine how many employees work in specific departments and to identify which machines or devices belong to employees I particular office locations.

This project demonstrates practical use of SQL for filtering, pattern matching, and analysing organizational data in s secure, simulated environment.

Task 1: List all Organization machines and their operating system

Objective:
To retrieve and display a complete list of all devices within the organization along with their operating system information.

```
MariaDB [organization]> select device_id, operating_system
    -> from machines;
+-------------+------------------+
| device_id   | operating_system |
+-------------+------------------+
| a184b775c707 | OS 1            |
| a192b174c940 | OS 2            |
| a305b818c708 | OS 3            |
| a317b635c465 | OS 1            |
| a320b137c219 | OS 2            |
| a398b471c573 | OS 3            |
| a667b270c984 | OS 1            |
| a821b452c176 | OS 2            |
```

Command: Select device_id, operating_system
from machines;

Explanation: This query retrieves the device_id and operating_system columns from the machines table, effectively listing all devices in the organization along with their operating system.

Task 2: Identify How Many Devices Are Running OS 2 For Emergency Update

Objective: To filter the devices in the organization's database and identify those running OS 2, which requires an urgent update. This task demonstrates the use of the WHERE clause to retrieve specific records based on condition, supporting proactive device management and security maintenance.

```
MariaDB [organization]> select device_id, operating_system
    -> from machines
    -> where operating_system = 'OS 2';
+--------------+------------------+
| device_id    | operating_system |
+--------------+------------------+
| a192b174c940 | OS 2             |
| a320b137c219 | OS 2             |
| a821b452c176 | OS 2             |
| b157c491d493 | OS 2             |
| b264c773d977 | OS 2             |
| b265c937d713 | OS 2             |
| b806c503d354 | OS 2             |
| b979c871d361 | OS 2             |
| c150d982e144 | OS 2             |
```

```
| y347z204a710 | OS 2             |
| y765z123a548 | OS 2             |
| y976z753a267 | OS 2             |
| z451a308b518 | OS 2             |
| z654a154b259 | OS 2             |
+--------------+------------------+
80 rows in set (0.001 sec)
```

Command: Select device_id, operating_system
From machines
Where operating_system = 'OS 2'.

Explanation: From the results, we find that there are 80 devices running OS 2, highlighting the scope of the emergency update needed. This targeted approach is essential for efficient maintenance and reading potential security risks associated with outdated operating systems.

Task 3 Retrieve Employee Records in Finance Department and Their Office Numbers

Objective: To filter the employee records in the organization's database and identify all employees working in the Finance and along with their office numbers.

```
MariaDB [organization]> Select *
    -> from employees
    -> where department = 'finance';
+-------------+---------------+----------+------------+-------------+
| employee_id | device_id     | username | department | office      |
+-------------+---------------+----------+------------+-------------+
|        1003 | d394e816f943  | sgilmore | Finance    | South-153   |
|        1007 | h174i497j413  | wjaffrey | Finance    | North-406   |
|        1008 | i858j583k571  | abernard | Finance    | South-170   |
|        1010 | k2421212m542  | jlansky  | Finance    | South-109   |
|        1015 | p611q262r945  | jsoto    | Finance    | North-271   |
|        1017 | r550s824t230  | jclark   | Finance    | North-188   |
```

Command: Select*
From employees
Where department = 'finance';

Explanation: By using the WHERE clause, it filters the results to include only employees who work in the Finance department. We can see the employees working in the finance department and their office numbers

Task 4: Identify Employee Using the Unresponsive Machine in the South-109 Building

Objective: To determine which employee is assigned to the machine located in the South-109 building that has reported an issue.

```
MariaDB [organization]> select *
    -> from employees
    -> where office = 'south-109';
+-------------+-------------+----------+------------+-----------+
| employee_id | device_id   | username | department | office    |
+-------------+-------------+----------+------------+-----------+
|        1010 | k2421212m542 | jlansky | Finance    | South-109 |
+-------------+-------------+----------+------------+-----------+
```

Command: select *
From employees
Where office = ‘south-109’;

Explanation: This task involved combining data from the machines and employees table to identify which employee was linked to the unresponsive system in the South-109 building

Findings:
The query results showed that the computer in office South-109 is assigned to **employee ID 1010**,
With the user **jlansky**, who works in the **Finance Department**.

Task 5: Identify All Machines Located in the South Building

Objective: To identify all machines located in the South building, as the IT team has determined there are issues affecting all computer systems in that location

```
MariaDB [organization]> select *
    -> from employees
    -> where office like 'south%';
+-------------+---------------+----------+------------------------+-----------+
| employee_id | device_id     | username | department             | office    |
+-------------+---------------+----------+------------------------+-----------+
|        1003 | d394e816f943  | sgilmore | Finance                | South-153 |
|        1004 | e218f877g788  | eraab    | Human Resources        | South-127 |
|        1005 | f551g340h864  | gesparza | Human Resources        | South-366 |
|        1008 | i858j583k571  | abernard | Finance                | South-170 |
|        1009 | NULL          | lrodriqu | Sales                  | South-134 |
|        1010 | k2421212m542  | jlansky  | Finance                | South-109 |
|        1011 | 1748m120n401  | drosas   | Sales                  | South-292 |
|        1013 | n2050559p243  | zbernal  | Information Technology | South-229 |
|        1020 | u899v381w363  | arutley  | Marketing              | South-351 |
|        1021 | v200w121x977  | smartell | Information Technology | South-138 |
```

Command: select *
From employees
Where office like ‘south%’;

Explanation: To locate the only machines in the South building, a SQL query was used with the LIKE operator and the % wildcard – for example, matching office names that begin with “south”. The % symbol acts as a placeholder for any sequence of characters that follows, allowing the query to find every record where the office name starts with “south”, regardless of the specific room number.

The query allowed all the identified machines were isolated for troubleshooting and detailed inspection by the IT team

Conclusion/What I Learned

In this project, I learned how to use SQL to retrieve and filter data from an organization’s database to investigate and solve real- world IT issues. The project focused on using key filtering commands to refine query results and identify affected systems efficiently.

Key SQL commands Learned

- WHERE clause – Used to filter results based on specific conditions, such as finding employees in certain departments or machines located in specific buildings.
- LIKE operator – Used to search for patterns in data, for example, locating all offices beginning with “South” to identify affected devices.

Practical Outcomes

- Identified devices matching specific conditions (e.g., building or OS)
- Determined which employees were using affected computers.
- Isolated all machines located in the South building for troubleshooting and inspection.

This project enhanced my understanding of how SQL can be used to query, filter, and analyse organizational data effectively for IT system management and issue resolution.